Answer key. Answers are in magenta.

NRES 746, Week 6, Lecture 2: Find the Bugs

Instructions: Work alone for about 3 minutes, then compare with a neighbor. Keep laptops closed.

The code

Here are two short functions, possibly written with AI help. The first comes from today's live-coding problem (finishing Lab 4's toad monitoring design).

- `detect_prob(k)`: one person-day of searching detects an animal with probability `pobs`, so `k` person-days should detect it with probability $1 - (1 - p_{obs})^k$.
- `sim_detect(R)`: should simulate `R` surveys of a site where the frog from Lab 5 is present (3 visits per survey, detection probability 0.4 per visit) and return the fraction of surveys with at least one detection.

Each function below contains **exactly one bug**. Both run without an error, but give wrong answers.

```
detect_prob <- function(k, pobs = 0.02) {
  1 - pobs^k
}

sim_detect <- function(R = 1000, visits = 3, p = 0.4) {
  detected <- logical(R)
  for (i in seq_len(R)) {
    n_det <- rbinom(1, size = visits, prob = p)
    detected <- n_det > 0
  }
  mean(detected)
}
```

Part 1: Find and fix (individual, ~3 min)

For each function, circle the bug on the code above, write the corrected line, and say what would go wrong if you ran the buggy version.

1a. `detect_prob()`

Fix: `1 - (1 - pobs)^k`. The buggy version, $1 - p_{obs}^k$, returns nearly 1 for any effort (e.g. 0.9996 for $k = 2$), so detection looks almost certain.

1b. `sim_detect()`

Fix: `detected[i] <- n_det > 0`. The buggy version overwrites the whole vector every survey, so it returns only the last survey's result (exactly 0 or 1).

Part 2: Catch it next time (with a neighbor, ~2 min)

2. For one of the bugs, what quick check (one line of code and the answer you'd expect) would have caught it?

E.g. `detect_prob(1)` should return 0.02 (one day = `pobs`); or `sim_detect()` should be near $1 - 0.6^3 = 0.784$, never exactly 0 or 1.